# MSG No. 192.246 $P6'/mc'c$ [ Type III, hexagonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `6'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\, 0,\, \frac{1}{4}\right]$ | `[1,2,3,4,5,6,13,14,15,16,17,18]` |
| 2 | $\left[0,\, 0,\, \frac{3}{4}\right]$ | `[7,8,9,10,11,12,19,20,21,22,23,24]` |

Table 2: Wyckoff site: `2b`, site symmetry: `6'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\, 0,\, 0]$ | `[1,2,3,7,8,9,13,14,15,19,20,21]` |
| 2 | $\left[0,\, 0,\, \frac{1}{2}\right]$ | `[4,5,6,10,11,12,16,17,18,22,23,24]` |

Table 3: Wyckoff site: `4c`, site symmetry: `3.2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$ | `[1,2,3,16,17,18]` |
| 2 | $\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$ | `[4,5,6,13,14,15]` |
| 3 | $\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$ | `[7,8,9,22,23,24]` |
| 4 | $\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$ | `[10,11,12,19,20,21]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-6..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$ | `[1,2,3,7,8,9]` |
| 2 | $\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$ | `[4,5,6,10,11,12]` |
| 3 | $\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$ | `[13,14,15,19,20,21]` |
| 4 | $\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$ | `[16,17,18,22,23,24]` |

Table 5: Wyckoff site: `4e`, site symmetry: `6'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\, 0,\, z]$ | `[1,2,3,13,14,15]` |
| 2 | $\left[0,\, 0,\, \frac{1}{2} - z\right]$ | `[4,5,6,16,17,18]` |
| 3 | $[0,\, 0,\, -z]$ | `[7,8,9,19,20,21]` |
| 4 | $\left[0,\, 0,\, z + \frac{1}{2}\right]$ | `[10,11,12,22,23,24]` |

Table 6: Wyckoff site: `6f`, site symmetry: `2'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\, 0,\, \frac{1}{4}\right]$ | `[1,4,14,17]` |
| 2 | $\left[0,\, \frac{1}{2},\, \frac{1}{4}\right]$ | `[2,5,15,18]` |
| 3 | $\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$ | `[3,6,13,16]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$ | [7,10,21,24] |
| 5 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [8,11,19,22] |
| 6 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [9,12,20,23] |

Table 7: Wyckoff site: 6g, site symmetry: 2'/m..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | [1,8,14,19] |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | [2,9,15,20] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [3,7,13,21] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [4,11,17,22] |
| 5 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [5,12,18,23] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [6,10,16,24] |

Table 8: Wyckoff site: 8h, site symmetry: 3..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, z\right]$ | [1,2,3] |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$ | [4,5,6] |
| 3 | $\left[\frac{1}{3}, \frac{2}{3}, -z\right]$ | [7,8,9] |
| 4 | $\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$ | [10,11,12] |
| 5 | $\left[\frac{2}{3}, \frac{1}{3}, z\right]$ | [13,14,15] |
| 6 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$ | [16,17,18] |
| 7 | $\left[\frac{2}{3}, \frac{1}{3}, -z\right]$ | [19,20,21] |
| 8 | $\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$ | [22,23,24] |

Table 9: Wyckoff site: 12i, site symmetry: 2'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, z\right]$ | [1,14] |
| 2 | $\left[0, \frac{1}{2}, z\right]$ | [2,15] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | [3,13] |
| 4 | $\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$ | [4,17] |
| 5 | $\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$ | [5,18] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$ | [6,16] |
| 7 | $\left[\frac{1}{2}, \frac{1}{2}, -z\right]$ | [7,21] |
| 8 | $\left[\frac{1}{2}, 0, -z\right]$ | [8,19] |
| 9 | $\left[0, \frac{1}{2}, -z\right]$ | [9,20] |
| 10 | $\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$ | [10,24] |
| 11 | $\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$ | [11,22] |
| 12 | $\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$ | [12,23] |

Table 10: Wyckoff site: `12j`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{4}\right]$ | `[1,4]` |
| 2 | $\left[0,\ x,\ \frac{1}{4}\right]$ | `[2,5]` |
| 3 | $\left[-x,\ -x,\ \frac{1}{4}\right]$ | `[3,6]` |
| 4 | $\left[-x,\ -x,\ \frac{3}{4}\right]$ | `[7,10]` |
| 5 | $\left[x,\ 0,\ \frac{3}{4}\right]$ | `[8,11]` |
| 6 | $\left[0,\ x,\ \frac{3}{4}\right]$ | `[9,12]` |
| 7 | $\left[x,\ x,\ \frac{1}{4}\right]$ | `[13,16]` |
| 8 | $\left[-x,\ 0,\ \frac{1}{4}\right]$ | `[14,17]` |
| 9 | $\left[0,\ -x,\ \frac{1}{4}\right]$ | `[15,18]` |
| 10 | $\left[-x,\ 0,\ \frac{3}{4}\right]$ | `[19,22]` |
| 11 | $\left[0,\ -x,\ \frac{3}{4}\right]$ | `[20,23]` |
| 12 | $\left[x,\ x,\ \frac{3}{4}\right]$ | `[21,24]` |

Table 11: Wyckoff site: `12k`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 2x,\ \frac{1}{4}\right]$ | `[1,17]` |
| 2 | $\left[-2x,\ -x,\ \frac{1}{4}\right]$ | `[2,18]` |
| 3 | $\left[x,\ -x,\ \frac{1}{4}\right]$ | `[3,16]` |
| 4 | $\left[-x,\ -2x,\ \frac{1}{4}\right]$ | `[4,14]` |
| 5 | $\left[2x,\ x,\ \frac{1}{4}\right]$ | `[5,15]` |
| 6 | $\left[-x,\ x,\ \frac{1}{4}\right]$ | `[6,13]` |
| 7 | $\left[x,\ -x,\ \frac{3}{4}\right]$ | `[7,24]` |
| 8 | $\left[x,\ 2x,\ \frac{3}{4}\right]$ | `[8,22]` |
| 9 | $\left[-2x,\ -x,\ \frac{3}{4}\right]$ | `[9,23]` |
| 10 | $\left[-x,\ x,\ \frac{3}{4}\right]$ | `[10,21]` |
| 11 | $\left[-x,\ -2x,\ \frac{3}{4}\right]$ | `[11,19]` |
| 12 | $\left[2x,\ x,\ \frac{3}{4}\right]$ | `[12,20]` |

Table 12: Wyckoff site: `12l`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,8]` |
| 2 | $[-y,\ x-y,\ 0]$ | `[2,9]` |
| 3 | $[-x+y,\ -x,\ 0]$ | `[3,7]` |
| 4 | $\left[x-y,\ -y,\ \frac{1}{2}\right]$ | `[4,11]` |
| 5 | $\left[y,\ x,\ \frac{1}{2}\right]$ | `[5,12]` |
| 6 | $\left[-x,\ -x+y,\ \frac{1}{2}\right]$ | `[6,10]` |
| 7 | $[x-y,\ x,\ 0]$ | `[13,21]` |
| 8 | $[-x,\ -y,\ 0]$ | `[14,19]` |
| 9 | $[y,\ -x+y,\ 0]$ | `[15,20]` |
| 10 | $\left[x,\ x-y,\ \frac{1}{2}\right]$ | `[16,24]` |
| 11 | $\left[-x+y,\ y,\ \frac{1}{2}\right]$ | `[17,22]` |
| 12 | $\left[-y,\ -x,\ \frac{1}{2}\right]$ | `[18,23]` |

Table 13: Wyckoff site: `24m`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[-y,\ x-y,\ z]$ | `[2]` |
| 3 | $[-x+y,\ -x,\ z]$ | `[3]` |
| 4 | $\left[x-y,\ -y,\ \frac{1}{2}-z\right]$ | `[4]` |
| 5 | $\left[y,\ x,\ \frac{1}{2}-z\right]$ | `[5]` |
| 6 | $\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$ | `[6]` |
| 7 | $[-x+y,\ -x,\ -z]$ | `[7]` |
| 8 | $[x,\ y,\ -z]$ | `[8]` |
| 9 | $[-y,\ x-y,\ -z]$ | `[9]` |
| 10 | $\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$ | `[10]` |
| 11 | $\left[x-y,\ -y,\ z+\frac{1}{2}\right]$ | `[11]` |
| 12 | $\left[y,\ x,\ z+\frac{1}{2}\right]$ | `[12]` |
| 13 | $[x-y,\ x,\ z]$ | `[13]` |
| 14 | $[-x,\ -y,\ z]$ | `[14]` |
| 15 | $[y,\ -x+y,\ z]$ | `[15]` |
| 16 | $\left[x,\ x-y,\ \frac{1}{2}-z\right]$ | `[16]` |
| 17 | $\left[-x+y,\ y,\ \frac{1}{2}-z\right]$ | `[17]` |
| 18 | $\left[-y,\ -x,\ \frac{1}{2}-z\right]$ | `[18]` |
| 19 | $[-x,\ -y,\ -z]$ | `[19]` |
| 20 | $[y,\ -x+y,\ -z]$ | `[20]` |
| 21 | $[x-y,\ x,\ -z]$ | `[21]` |
| 22 | $\left[-x+y,\ y,\ z+\frac{1}{2}\right]$ | `[22]` |
| 23 | $\left[-y,\ -x,\ z+\frac{1}{2}\right]$ | `[23]` |
| 24 | $\left[x,\ x-y,\ z+\frac{1}{2}\right]$ | `[24]` |